BINARY SEARCH TREES

Problem Solving with Computers-II

Join iclicker at **https://join.iclicker.com/GEXG**

Problem Scenario: Building Leetcode

You are building a competitive coding platform (think LeetCode contests). During a live contest, three things happen constantly:

- A student submits a solution and their **score updates** (insert/update)
- The UI needs to display the leaderboard in ranked order at any time
- A student wants to know "who is ranked just above and below me?" (predecessor/successor)

Discuss: Why does a sorted array of scores fall short in this case?

Problem Scenario: Task Scheduler

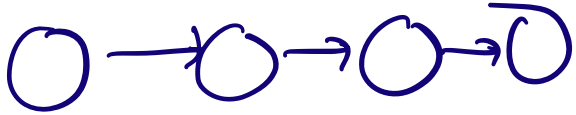
Your laptop runs Spotify, Chrome, and a virus scan simultaneously
The CPU can only run one task at a time.

Rule: run whichever task has received the least CPU time.

Thousands of times per second:
- Find the most starved task (lowest CPU time usage)
- Run it briefly
- Update its total, insert it back

Discuss: Why does a sorted array fail when a task's CPU time usage changes every few milliseconds?

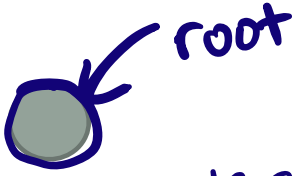
Which of the following is/are a tree?



Empty tree

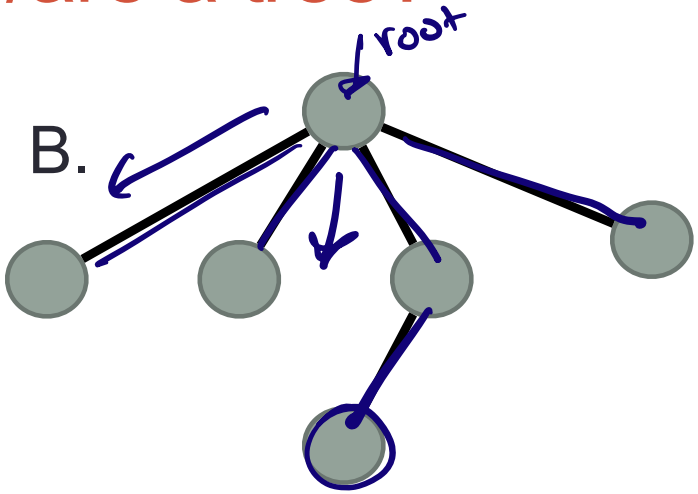
root

A.

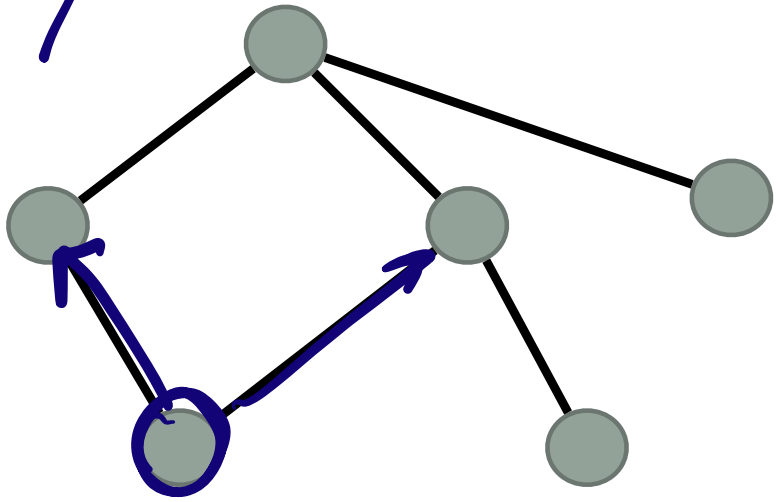


one-node tree

B.



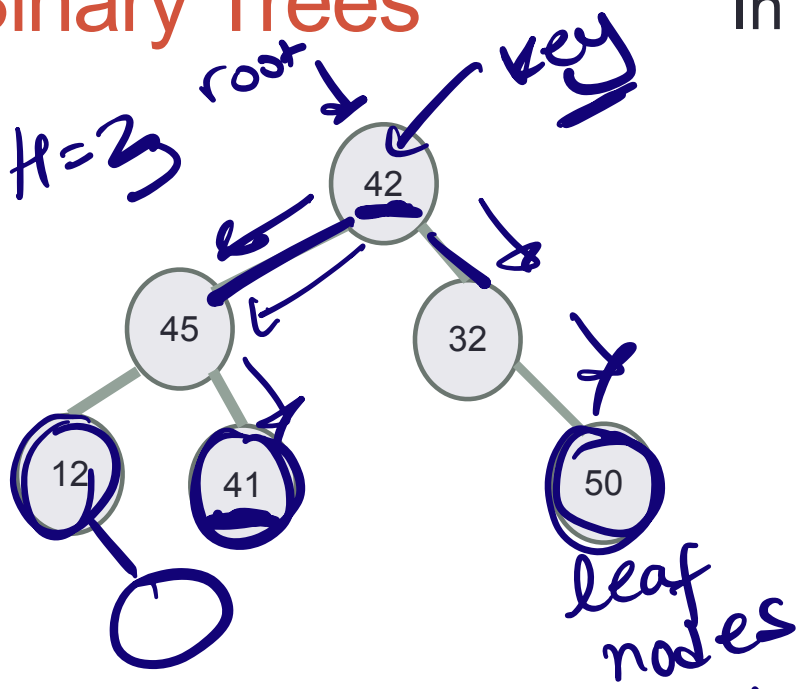
C.



D. A & B

E. All of A-C

Binary Trees



In a tree, nodes are arranged in a hierarchy

- One node is distinguished as the root
- Each node:
  - stores a key
  - has a pointer to child nodes and parent (optional)
- Unique path between any two nodes
- Leaf nodes have no children

Path: 42 → 45 → 41

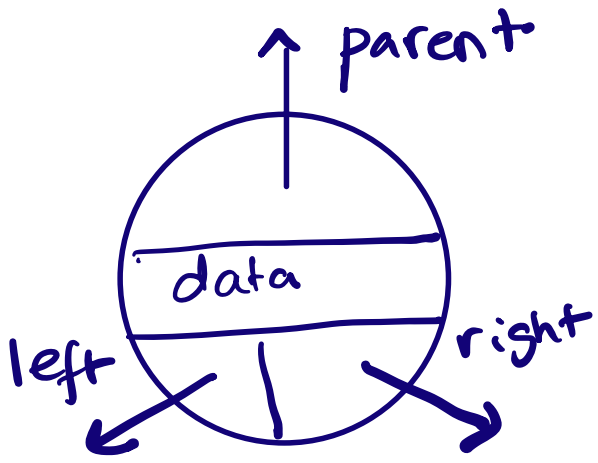
Height of the tree: length of the longest path from root to leaf

In a binary tree, each node has at most two children

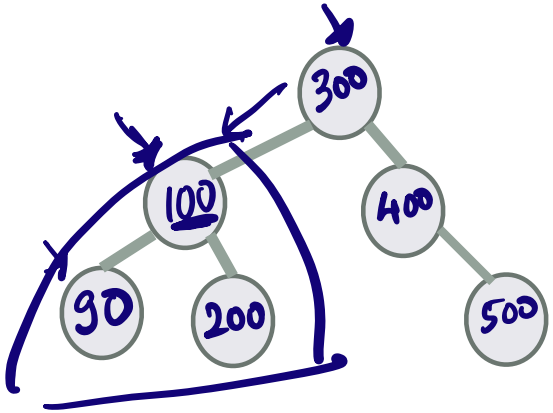
```
struct TreeNode {

  TreeNode* left;
  TreeNode* right;
  TreeNode* parent;
  int const data;

  TreeNode(int d) : data(d) {
    left = right = parent = nullptr;
  }
};
```

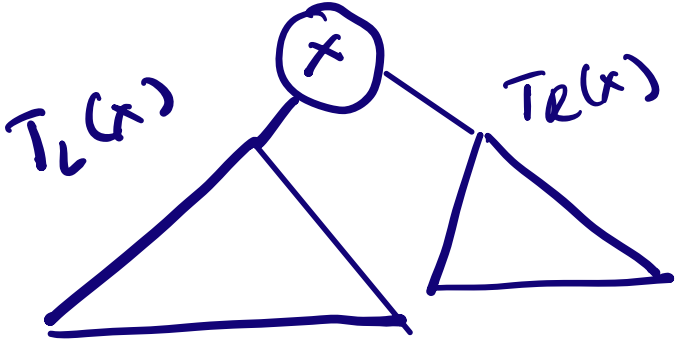


Binary Search Tree – What is it?



BST is a binary tree where each node satisfies the Search Tree Property

keys in $T_L(x)$ < key (x) ≤ keys in $T_R(x)$



$T_L(x)$: Left subtree of x

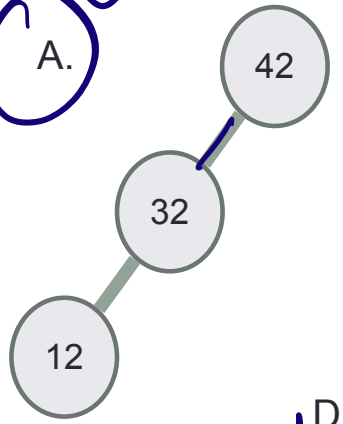
$T_R(x)$: Right subtree of x

Do the keys have to be integers?

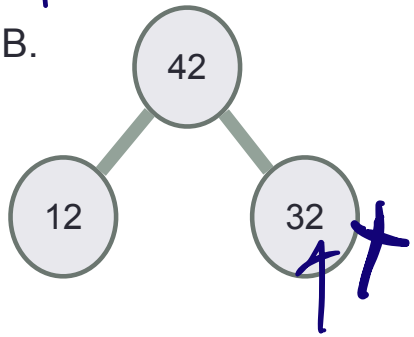
Visualization: https://visualgo.net/en/bst

Which of the following is/are a binary search tree?

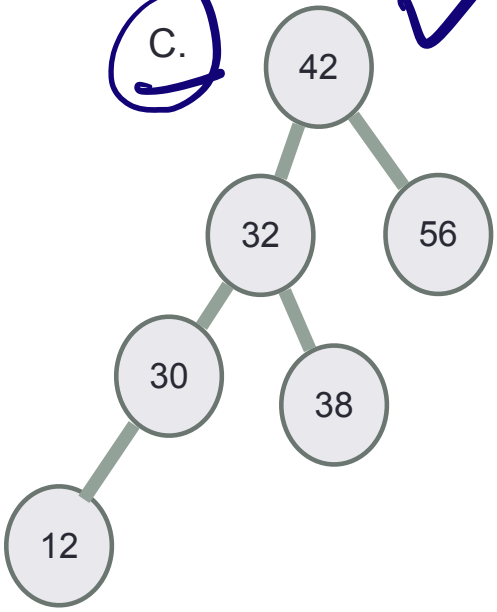
A.



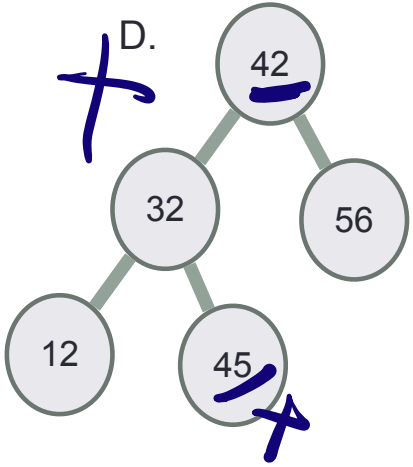
B.



C.

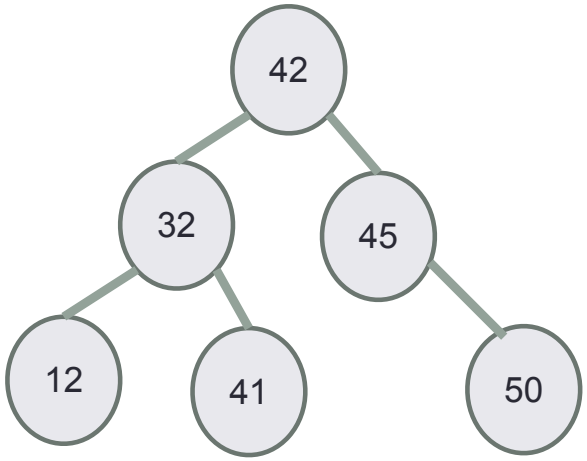


D.



E. More than one of these

Search



- Start at the root;
- Trace down a path by comparing **k** with the key of the current node x:
    - If the keys are equal: we have found the key
    - If **k** < key[x] search in the left subtree of x
    - If **k** > key[x] search in the right subtree of x
- What is the running time of search?

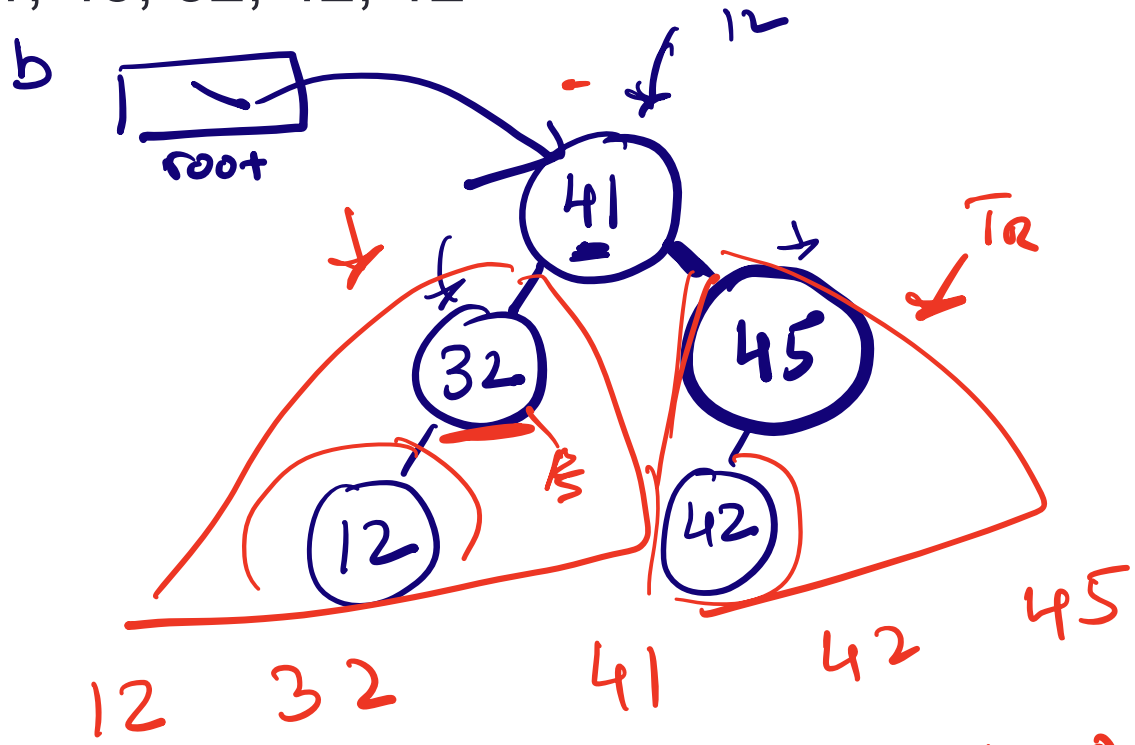
Search for 41, then search for 53

Operation: Insert

Insert keys: 41, 45, 32, 42, 12

bst b;

b.insert(41)
b.insert(45)
b.insert(32)
b.insert(42)
b.insert(12)

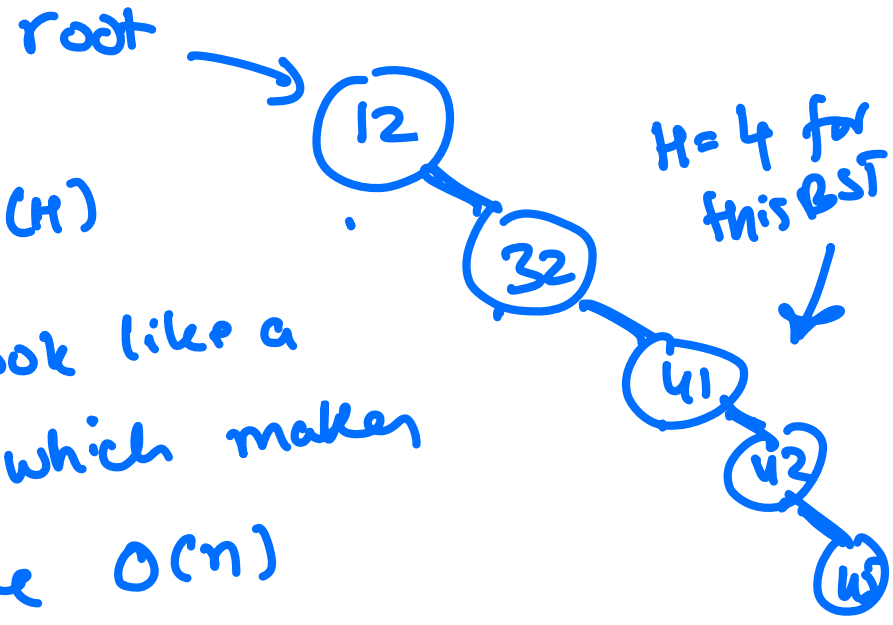


In a balanced tree, H = O(log n). Worst case running time for insert is O(H) = O(log n)

Insert key in sorted order: 12, 32, 41, 42, 45
What is the height of the tree?
What is the best/worst case running time for insert?

Worst case running time for search, insert (and many other BST operations) is $O(H)$

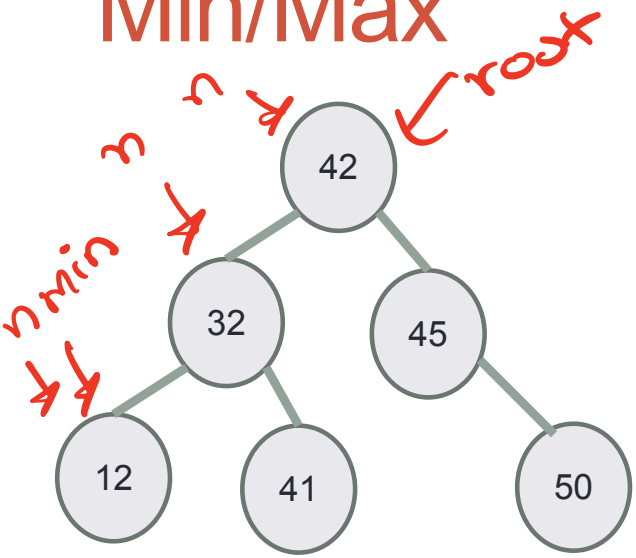
If the BST is skewed to look like a linked list, then $H = n-1$, which makes the worst case running time $O(n)$



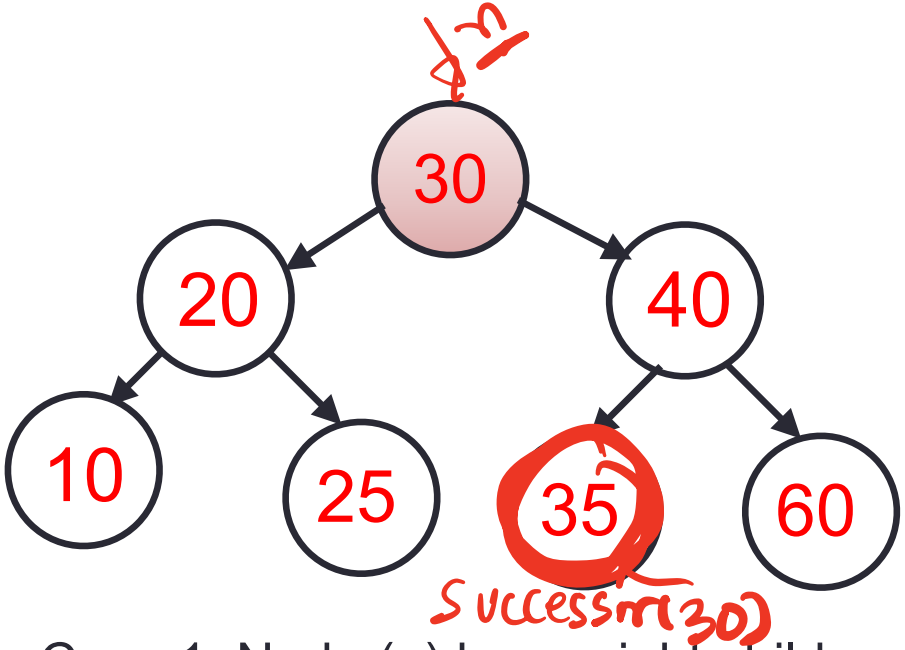
H = 4 for this BST

H: height of any BST

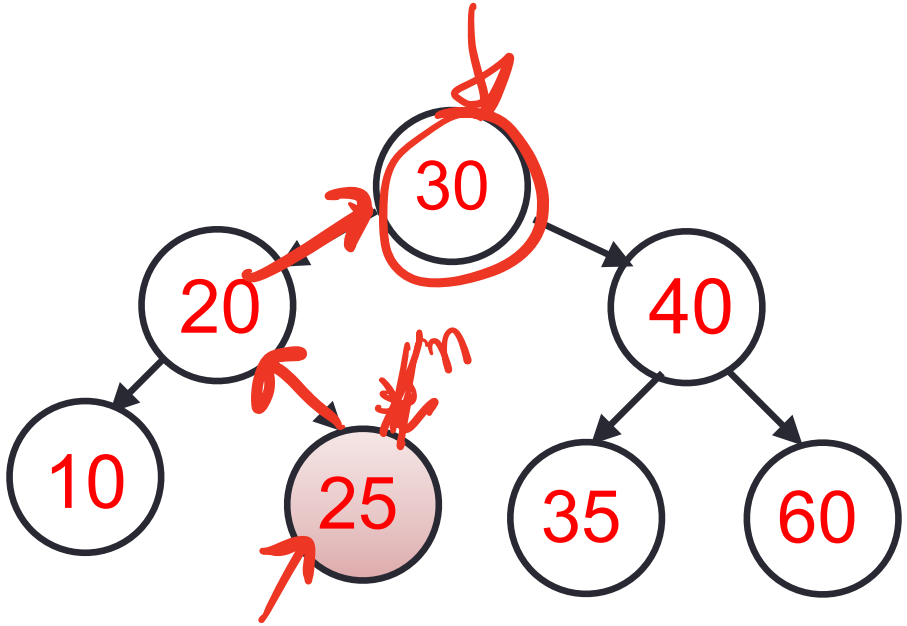
Min/Max



Successor: Next largest element



- Case 1: Node (n) has a right child

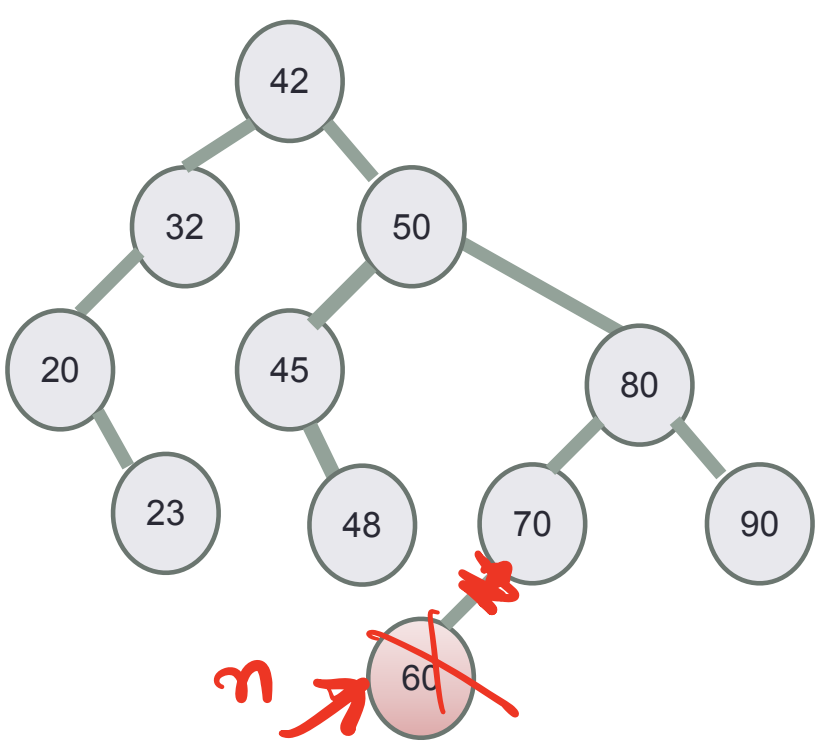


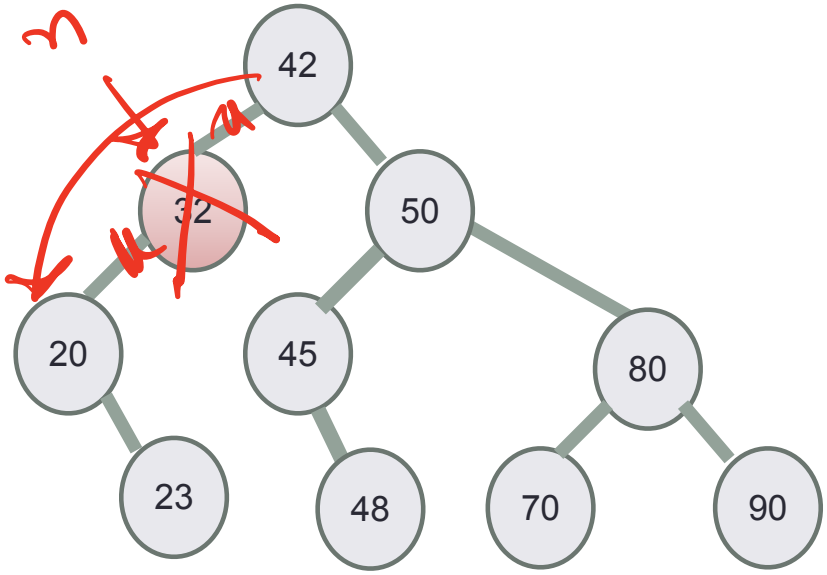
- Case 2: Node (n) has no right child

successor(n) is the min key in $T_R(n)$

Delete a specific node

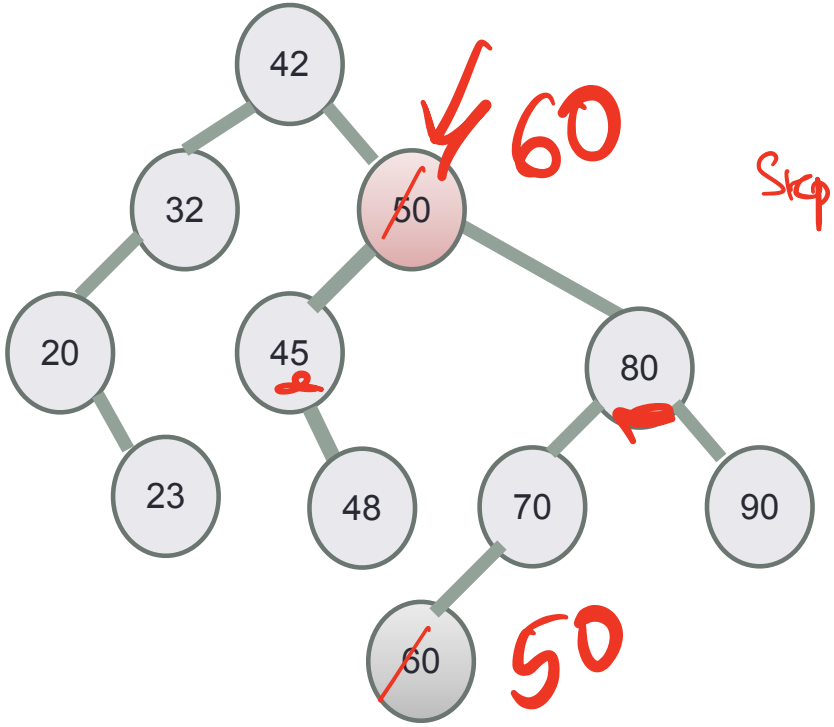
delete (32





- Case 1: Node is a leaf node
  - Set parent's (left/right) child pointer to null
  - Delete the node
- Case 2: Node has one child (left/right)
  - Replace the node by its only child

Delete a specific node



Example delete(50)

- **Case 3: Node (x) has two children**

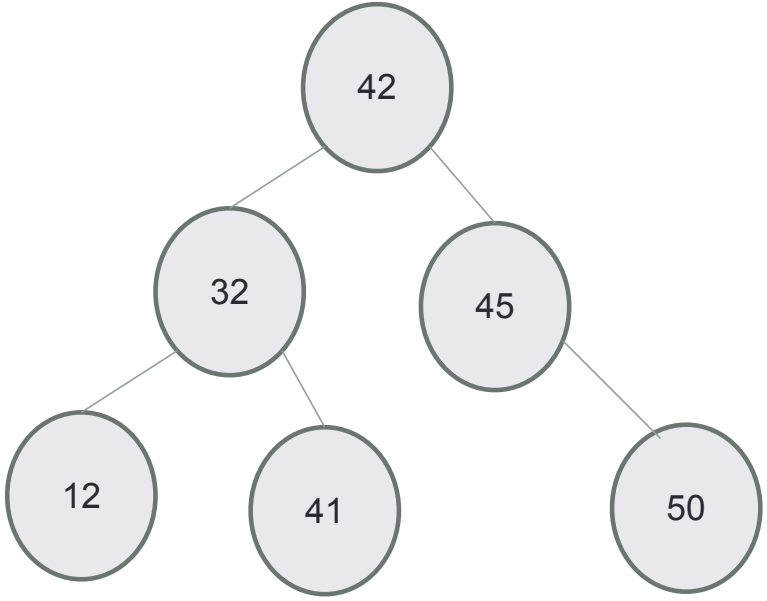
Step 1: Swap key(x) with successor(x) (or predecessor(x).)

In this example x = 50, so we swap 50 with 60 (successor(50)).

Since the node has two children, (1) successor is guaranteed to be in the right subtree. (2) successor(x) has no left child (min node in right subtree)

Therefore we can delete 50 at the new position using case (1) or (2) of delete.

Step 2: Delete the node(x) at the position of its successor using case (1) or (2)

BST ADT

Operations
Search
Insert
Min
Max
Successor
Predecessor
Delete
Print elements In order Preorder, Post order